

SPRINT 01

Telemetry & Ingestion Contract Approval Report

Industrial Equipment Monitoring Platform

Document	Telemetry & Ingestion Contract Approval Report
Milestone	M1 - Contracts and Vertical Slice
Role	Technical Lead
Contract	telemetry-batch.v1 / schema_version 1
Baseline	Requirements Specification v1.0 - Draft baseline
Decision status	APPROVED FOR IMPLEMENTATION - review acknowledgements tracked
Report date	19 September 2026

Purpose

Approve the versioned telemetry and ingestion contract used by firmware/gateway simulators, the Raspberry Pi gateway, FastAPI ingestion, PostgreSQL persistence and QA tests. The approval freezes identity, timestamp ownership, authenticated gateway identity, per-item outcomes, error semantics, batch limits and commit behavior before reliability implementation expands.

Technical Lead decision package - contract-first, testable and traceable

1. Executive Approval Decision

Technical Lead approval

The telemetry and ingestion contract defined in this report is approved as the Sprint 01 implementation baseline. It is suitable for backend, gateway and QA implementation because event identity, timestamp ownership, gateway authentication, item-level outcomes, error codes, batch limits and database commit semantics are deterministic. Repository merge/release evidence remains incomplete until Backend Lead and Gateway Developer acknowledgements are recorded in the review register.

This decision is grounded in the project requirements, the QA Sprint 1 contract review, and the existing Sprint 01 Technical Lead report. The source baseline requires stable event identity, gateway-side durable replay semantics, idempotent backend ingestion, per-item mixed-batch outcomes, uniqueness on (device_id, boot_id, sequence_number), maximum batch size 500, and acceptance only after database commit.

1.1 Approval scope

- Versioned telemetry batch payload and required fields.
- Stable telemetry event identity and duplicate/conflict semantics.
- Device, gateway and backend timestamp ownership plus clock-quality semantics.
- Authenticated gateway identity derived from credentials, not from caller-supplied site data.
- Per-item accepted / duplicate / rejected outcomes and stable machine-readable errors.
- Mixed-batch behavior, maximum batch size of 500, and database commit boundary.
- Reference fixtures for valid, matching duplicate, conflicting duplicate, invalid, mixed and oversized cases.

1.2 Source basis

Source	Contract evidence used
Industrial Equipment Monitoring Platform - Requirements Specification v1.0	Sections 4-6, 8, 10 and 12: implementation baseline; FR-05 to FR-09; telemetry fields and uniqueness; commit and queue rules; batch size 500; AT-05/AT-06; ADR deliverable expectations.
QA Sprint 1 - Telemetry Contract, Decision Log and Open Questions	QA-D01 event identity; QA-D02 timestamp separation; QA-D03 mixed per-item outcomes; duplicate/conflict tests; lost-response-after-commit scenario; open questions on comparison fields, HTTP semantics and rejection taxonomy.
Sprint 01 Technical Report - Technical Lead	ADR-001 identity, ADR-002 timestamp ownership, immutable content, proposed endpoint and outcome vocabulary, rejection codes, commit boundary, gateway queue lifecycle and technical decision register.

2. Contract Principles and Responsibility Boundaries

Principle	Approved rule
Versioned contract	The request envelope uses schema_version = 1. Contract changes that alter validation or semantics require a new version or explicitly backward-compatible extension.
Stable event identity	event_identity = (device_id, boot_id, sequence_number). Retries preserve all three values. A new boot receives a new boot_id.
Authenticated gateway	Gateway identity is derived from its credential. The backend verifies that each device belongs to that gateway. site_id in a payload is never proof of authorization.
Independent timestamps	Device measurement time, gateway receipt time and backend receipt time remain separate. All persisted absolute timestamps are UTC.
Idempotent storage	Matching retry produces duplicate; conflicting reuse of the same identity produces identity_conflict. Database uniqueness is authoritative.
Per-item processing	A syntactically valid mixed batch may contain accepted, duplicate and rejected results. Invalid items do not block unrelated valid items.
Commit boundary	No item is reported accepted until the PostgreSQL transaction containing that item has committed successfully.
Bounded batch	1 to 500 events are accepted per request. A request with more than 500 events is rejected before item processing.

Responsibility boundary

The backend decides the authoritative outcome of each event. The gateway maps that outcome to queue lifecycle: accepted/duplicate -> remove; permanent rejection -> quarantine; timeout/429/5xx -> keep queued and retry.

3. Approved Versioned Payload

3.1 Endpoint and request envelope

```
POST /api/v1/telemetry/batches
Content-Type: application/json
Authorization: Bearer <gateway-credential>
```

The credential authenticates the gateway. The body does not authorize itself. The backend resolves the registered gateway from the credential and verifies every event device against that gateway assignment.

```
{
  "schema_version": 1,
  "events": [
    {
      "device_id": "motor-001",
      "boot_id": "7a9678d2-3319-47e2-a5be-8c688978720b",
      "sequence_number": 1842,
      "measured_at": "2026-09-19T00:30:00Z",
      "device_uptime_ms": 534000,
      "gateway_received_at": "2026-09-19T00:30:01.120Z",
      "metric": "temperature",
      "value": 31.4,
      "unit": "celsius",
      "quality": {
        "reading": "valid",
        "clock": "synchronised"
      }
    }
  ]
}
```

3.2 Field contract

Field	Owner	Required / rule
schema_version	Contract	Required at batch envelope; v1 supports value 1. Unsupported versions are rejected at request level.
device_id	Device domain	Required registered device identifier. Ownership is checked against authenticated gateway.
boot_id	Device	Required identifier unique to the current device boot.
sequence_number	Device	Required non-negative integer, monotonically increasing within a boot.
measured_at	Device	UTC physical measurement time when trustworthy; otherwise null.
device_uptime_ms	Device	Elapsed milliseconds since current boot.
gateway_received_at	Gateway	UTC time of original gateway receipt. The gateway retains this value across retries.
backend_received_at	Backend	Server-generated only; never accepted from the client request.
metric	Contract	temperature for the MVP.
value	Device	Finite numeric temperature within the selected sensor range.
unit	Contract	celsius for the MVP.
quality.reading	Device	v1 accepted value: valid. Unknown values are rejected until a broader reading-quality/status contract is approved.
quality.clock	Device	Approved enum: synchronised, unsynchronised, estimated, unknown. Controls trust of measured_at for later history/live ordering.

Scope note

The requirements require an explicit disconnected-sensor error status, but the supplied source documents do not freeze the full invalid-reading payload representation. This approval therefore freezes the telemetry-reading ingestion contract and clock-quality vocabulary; a richer sensor-status/invalid-reading representation remains a separate M2 contract decision.

4. Stable Event Identity and Duplicate Protection

```
event_identity = (device_id, boot_id, sequence_number)
```

PostgreSQL domain constraint:

```
UNIQUE (device_id, boot_id, sequence_number)
```

The database constraint is authoritative. Application-side checks may improve diagnostics but cannot replace database uniqueness.

4.1 Approved duplicate comparison

For Sprint 01, the backend compares immutable telemetry fields directly. A canonical content hash is not required for the MVP. This resolves the QA duplicate-comparison question without adding another persistence field.

Included in immutable comparison	Excluded from immutable comparison
schema_version, device_id, boot_id, sequence_number, measured_at, device_uptime_ms, metric, value, unit, quality	gateway_received_at, backend_received_at, HTTP metadata, retry count, batch_id

Existing identity?	Incoming content	Outcome	Database action
No	Valid	accepted	Insert new row; respond only after commit.
Yes	Same immutable content	duplicate	Do not insert another row.
Yes	Different immutable content	rejected / identity_conflict	Keep original row unchanged; log conflict.

5. Timestamp and Quality Semantics

Timestamp	Owner	Meaning	Caller override
measured_at	Device	Physical measurement time when the device clock is trustworthy; may be null.	Device payload value
device_uptime_ms	Device	Elapsed time since current device boot; supports ordering/debug context.	Device payload value
gateway_received_at	Gateway	Original UTC time the gateway received the reading.	Not ESP32-supplied
backend_received_at	Backend	UTC time the backend receives/processes the event.	Never client-supplied

Approved clock-quality values are synchronised, unsynchronised, estimated and unknown. A trustworthy historical measurement time requires `quality.clock = synchronised`. Other values preserve `measured_at` if supplied but must not silently substitute `backend_received_at` as the physical measurement time.

```
"quality": {
  "reading": "valid",
  "clock": "synchronised"
}
```

6. Authenticated Gateway Identity and Ownership Validation

```

gateway credential
|
v
registered gateway
|
+--> device assignment check
|
+--> allowed: continue ingestion
+--> not assigned: rejected / wrong_gateway

```

The request payload does not need `gateway_id` or `site_id` as authoritative fields. A caller-supplied site identifier cannot prove access. This aligns with the baseline rule that backend authorization derives from the registered gateway/device relationship.

Condition	Handling
Missing/invalid/revoked gateway credential	Reject request before item processing (401/403 according to authentication implementation). No accepted outcomes are returned.
Authenticated gateway, unknown device_id	Per-item rejected with <code>unknown_device</code> .
Authenticated gateway, device belongs to another gateway	Per-item rejected with <code>wrong_gateway</code> .
Authenticated gateway, assigned device	Continue schema, duplicate and persistence processing.

7. Ingestion Processing, Per-item Outcomes and Errors

7.1 Processing sequence

```

Receive request
-> authenticate gateway credential
-> validate envelope and batch size (1..500)
-> validate each item
-> verify device ownership
-> classify permanent rejections
-> detect duplicates / identity conflicts
-> stage inserts for valid new events
-> COMMIT PostgreSQL transaction
-> only now mark staged items as accepted
-> return per-item results

```

Non-negotiable commit rule

Never report an item as accepted before the PostgreSQL transaction has committed. If the commit fails or the server cannot establish the commit outcome before responding, return a transient request failure and allow the gateway to retry. Idempotency makes a retry after an actually successful commit safe.

7.2 Per-item outcome vocabulary

Outcome	Meaning	Gateway action
accepted	New valid event persisted; database commit succeeded.	Remove from delivery queue.
duplicate	Same event identity and same immutable content already stored.	Remove from delivery queue.
rejected	Permanent item-level failure with stable reason code.	Move to bounded quarantine; remove from delivery queue.

7.3 Approved error taxonomy

Code	Scope	Meaning
<code>unknown_device</code>	Item	<code>device_id</code> is not registered.

Code	Scope	Meaning
wrong_gateway	Item	Authenticated gateway does not own the device.
invalid_metric	Item	metric is not supported by v1.
invalid_unit	Item	unit is not celsius.
value_out_of_range	Item	Finite numeric value is outside the selected sensor range.
malformed_value	Item	Value or required item data is malformed.
identity_conflict	Item	Existing identity is reused with different immutable content.
unsupported_schema_version	Request	schema_version is not supported.
batch_too_large	Request	events contains more than 500 items.
malformed_batch	Request	Envelope is syntactically invalid or events is absent/not an array.

7.4 HTTP semantics

HTTP result	Approved use
200 OK	Syntactically valid v1 request completed and database commit succeeded. Body contains one deterministic result per input item, including mixed accepted/duplicate/rejected outcomes.
400 Bad Request	Malformed batch, unsupported schema version, empty batch if disallowed by API, or batch_too_large. No item is accepted.
401 / 403	Gateway credential is missing, invalid, revoked or not authorized. No item is accepted.
429 Too Many Requests	Transient request-level failure. Gateway keeps unconfirmed items queued and retries with bounded backoff.
5xx	Transient server/database failure. No accepted claims are returned for the failed request; gateway retries.

7.5 Response envelope

```
{
  "batch_id": "01K5EXAMPLE",
  "results": [
    {
      "device_id": "motor-001",
      "boot_id": "boot-a",
      "sequence_number": 100,
      "outcome": "accepted"
    },
    {
      "device_id": "motor-001",
      "boot_id": "boot-a",
      "sequence_number": 101,
      "outcome": "rejected",
      "reason": "invalid_unit"
    }
  ]
}
```

8. Approved Example Fixtures

The following fixtures are normative examples for Backend, Gateway and QA tests. They are intentionally small so expected behavior can be predicted without implementation-specific knowledge.

Fixture F01 - Valid event

```
REQUEST (schema_version 1)
{
  "device_id": "motor-001",
  "boot_id": "boot-a",
  "sequence_number": 100,
  "measured_at": "2026-09-19T00:30:00Z",
  "device_uptime_ms": 534000,
  "gateway_received_at": "2026-09-19T00:30:01Z",
  "metric": "temperature",
  "value": 31.4,
  "unit": "celsius",
  "quality": {"reading": "valid", "clock": "synchronised"}
}
EXPECTED: accepted, exactly one stored row, response only after commit.
```

Fixture F02 - Matching duplicate

PRECONDITION: F01 is already committed.
 REQUEST: repeat F01 with the same event identity and same immutable content.
 EXPECTED: duplicate; telemetry row count remains 1; gateway may remove event from queue.

Fixture F03 - Conflicting duplicate

PRECONDITION: F01 is already committed.
 REQUEST: same device_id= motor-001, boot_id=boot-a, sequence_number=100,
 but value=32.1 instead of 31.4.
 EXPECTED: rejected, reason=identity_conflict; original row remains unchanged; conflict is logged.

Fixture F04 - Invalid event

REQUEST: a new identity with unit="fahrenheit".
 EXPECTED: rejected, reason=invalid_unit; no telemetry row is inserted for the invalid event.

Fixture F05 - Mixed batch

INPUT ITEMS
 101 -> new valid event
 100 -> exact retry of committed F01
 102 -> new event with unit="fahrenheit"
 103 -> new valid event

EXPECTED AFTER ONE SUCCESSFUL DATABASE COMMIT
 101 -> accepted
 100 -> duplicate
 102 -> rejected: invalid_unit
 103 -> accepted
 HTTP 200 with four item results. Invalid item 102 does not block 101 or 103.

Fixture F06 - Maximum batch boundary

500 events -> process normally subject to per-item validation.
 501 events -> HTTP 400, reason=batch_too_large; no item processing and no accepted outcomes.

Fixture F07 - Commit succeeds but response is interrupted

1. Gateway submits new valid event #200.
2. Backend inserts and COMMIT succeeds.
3. Network response is lost.
4. Gateway retries the exact same identity/content.
5. Backend returns duplicate.

EXPECTED: exactly one stored telemetry row; gateway can safely remove retry after duplicate response.

9. Acceptance Criteria Verification

Acceptance criterion	Evidence in this report	Status
Versioned payload defined	Section 3: telemetry-batch.v1, schema_version 1, field contract.	PASS
Stable event identity defined	Section 4: (device_id, boot_id, sequence_number) + PostgreSQL uniqueness.	PASS
Timestamps/quality defined	Section 5: timestamp ownership, UTC semantics and clock-quality enum.	PASS
Authenticated gateway identity defined	Section 6: gateway derived from credential; device ownership verified.	PASS
Per-item outcomes and errors defined	Section 7: accepted/duplicate/rejected and stable reason taxonomy.	PASS
Valid example provided	Fixture F01.	PASS
Matching duplicate example provided	Fixture F02.	PASS
Conflicting duplicate example provided	Fixture F03.	PASS
Invalid example provided	Fixture F04.	PASS
Maximum batch is 500	Section 2 and Fixture F06.	PASS
Mixed batches supported	Section 7 and Fixture F05.	PASS
Acceptance only after commit	Section 7.1 and Fixture F07.	PASS
QA review recorded	QA Sprint 1 working document is incorporated as review input; QA decisions/open questions are resolved or tracked below.	RECORDED
Backend review recorded	Review row created below; no completed Backend Lead acknowledgement is present in the supplied project evidence.	PENDING
Gateway review recorded	Review row created below; no completed Gateway Developer acknowledgement is present in the supplied project evidence.	PENDING

Acceptance conclusion

The technical contract itself satisfies the implementation and behavioral criteria. The collaboration criterion is not fully closed until Backend Lead and Gateway Developer acknowledgements are added to the repository review record. This report does not fabricate those approvals.

10. Review and Approval Record

Role	Review evidence / decision	Date	Status
Technical Lead	Contract decisions approved in this report; implementation baseline authorized.	19 Sep 2026	APPROVED
QA	QA Sprint 1 Telemetry Contract, Decision Log and Open Questions reviewed event identity, timestamps, mixed batches, duplicate/conflict and lost-response behavior. Inputs incorporated.	Sprint 01	REVIEW INPUT RECORDED
Backend Lead	Confirm endpoint schema, transaction strategy, uniqueness migration, reason codes and HTTP behavior.	_____	PENDING ACKNOWLEDGEMENT
Gateway Developer	Confirm gateway_received_at ownership, batch mapping, queue actions and transient retry behavior.	_____	PENDING ACKNOWLEDGEMENT

Recommended repository evidence: record role, reviewer name, date, commit/PR reference, disposition (approved / approved with comments / changes requested), and any follow-up issue IDs. A review is "recorded" only when that evidence is committed or attached to the implementation PR.

11. Decision Resolution and Remaining Open Items

11.1 QA/Technical questions resolved by this approval

Prior question	Approved resolution
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Prior question	Approved resolution
QA-OQ01 - clock quality values	synchronised / unsynchronised / estimated / unknown.
QA-OQ05 - duplicate comparison fields	Direct comparison of immutable telemetry fields listed in Section 4.1; no content hash required for MVP.
QA-OQ06 - HTTP semantics for mixed batch	HTTP 200 only after successful processing/commit; item-level accepted/duplicate/rejected results may be mixed.
QA-OQ07 - stable rejection taxonomy	Codes in Section 7.3 are the v1 contract baseline.

11.2 Explicitly not resolved by this ingestion approval

- Cross-boot live-alert ordering and how a new boot supersedes an old boot for live state.
- Exact alert stale-event age calculation and >15-second gap timeline.
- Full invalid-reading / disconnected-sensor status payload beyond the v1 valid telemetry-reading contract.
- Gateway quarantine capacity/retention and detailed retry/backoff timing parameters.
- Concurrent same-device alert-state serialization strategy; this is required before the full alert engine milestone.

12. Requirement Traceability

Requirement / scenario	Approved contract relevance
FR-05	Stable event identity; retries retain device_id, boot_id and sequence_number.
FR-06	Gateway receipt timestamp and queue-compatible outcome semantics support persistence-before-forwarding.
FR-07	Request-level transient failures remain retryable; idempotency makes retries safe.
FR-08	Database uniqueness, duplicate result and identity_conflict rule.
FR-09	Per-item validation and mixed accepted/duplicate/rejected results.
Telemetry processing rules	Accepted only after commit; gateway removes accepted/matching duplicate; invalid items quarantine; transient failures retry; max batch 500.
AT-05	Commit then lost response -> retry -> duplicate -> one stored row.
AT-06	Mixed valid, invalid and duplicate items receive individual deterministic outcomes.
M1	Versioned contract and simulator-to-API semantics are frozen before later gateway reliability implementation.

13. Technical Lead Approval Statement

APPROVED FOR IMPLEMENTATION

telemetry-batch.v1 is approved as the Sprint 01 telemetry and ingestion baseline. Backend, Gateway and QA implementations should use the fixtures and outcome semantics in this report as executable contract examples. No implementation may claim an event was accepted before its database commit succeeds. Matching retries must be idempotent, conflicting identity reuse must be rejected, and mixed batches must return deterministic per-item results. The contract is ready for implementation; the review gate closes when Backend Lead and Gateway Developer acknowledgements are recorded.

Technical Lead	Approved - 19 September 2026
Backend Lead	Pending acknowledgement: _____
Gateway Developer	Pending acknowledgement: _____

Source References

- Industrial Equipment Monitoring Platform - Requirements Specification, Version 1.0 Draft baseline: pp. 2-4 (implementation baseline, FR-05 to FR-09, telemetry contract and processing rules), p. 6 (AT-03 to AT-06), p. 7 (M1 and Definition of Done).
- QA Sprint 1 - Telemetry Contract, Decision Log and Open Questions: pp. 2-4 (identity, timestamps, batch outcomes and lost-response-after-commit), pp. 6-8 (decision register, verification matrix and traceability).
- Sprint 01 Technical Report - Technical Lead: pp. 4-6 (ADR-001 identity, ADR-002 timestamp ownership, immutable content, endpoint/outcomes, commit boundary and queue lifecycle), pp. 8-10 (decision register, open questions and exit checklist).